

Theo Blauberg

Senior Analytics Engineer · M.Sc. Economics · M.Sc. Data Science (ongoing)

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// PROFILE

Senior Analytics Engineer with an advanced degree in Economics and extensive experience in econometrics, machine learning, and experimentation. Currently completing a Master's in Data Science at the University of Helsinki. Proficient in R, Python, SQL, and GPU computing. Experienced in building production ML models, interactive data applications, and internal analytics tooling.

// EDUCATION

Master's Program in Data Science University of Helsinki

09/2021 -

- › **Major:** Data Science
- › Specializing in machine learning methods with emphasis on GPU-accelerated computing.
- › **Thesis:** Applying tabular foundation models to the detection of fraud patterns in high-dimensional transaction data. Work in progress; current draft.

Master's Program in Economics University of Helsinki

09/2020 - 12/2022

- › **Major:** Economics
- › **Minors:** Statistics, Mathematics, Computer Science
- › **Master's Thesis:** Impact of Policy Shocks in the EU Emissions Trading System on Finland.

Exchange Program Fudan University, Shanghai

09/2018 - 06/2019

Studied Chinese language, politics, and economics, with focus on East Asian economic development.

Language Course University of Vienna

09/2015 - 12/2015

Intensive German language studies.

// TECHNICAL SKILLS

Languages: R, Python, SQL, Julia, C++

Methods: causal inference, RCT analysis, time-series econometrics

ML/AI: Torch, tidymodels, scikit-learn, Claude Agent SDK, Anthropic Python SDK, Chatlas

Data: dplyr, DuckDB, pandas, Arrow, Spark, Hive, Impala, Snowflake

GPU: CUDA Python, RAPIDS cuDF

// WORK EXPERIENCE

Senior Analytics Engineer Nordea

07/2025 -

- › Build, maintain, and monitor machine learning models that deliver real-time fraud detection for card and account-to-account transactions.
- › Lead analytics initiatives with increased autonomy in model development and deployment decisions.
- › Maintain an internal R package for data analysis and visualization, significantly improving team productivity.

Data Analyst Nordea

07/2023 - 07/2025

- › Collaborated within a team responsible for building and monitoring real-time fraud detection models.
- › Developed internal R packages, Shiny apps, and automated reports. Established a Data & Analytics community within the department.

Data Analyst VATT Institute for Economic Research

01/2023 - 06/2023

- › Conducted analyses and co-authored research reports on electricity market data, focusing on [consumer](#) and [company](#) responses to price shocks.
- › Designed and developed a [Shiny-based dashboard](#) for interactive data visualization.

Research Assistant VATT Institute for Economic Research

08/2022 - 01/2023

- › Supported research projects with data preparation, analysis, and visualization tasks.

Project Worker City of Tampere

05/2022 - 08/2022

- › Analysed a randomised controlled experiment (~5,000 users) testing whether information nudges could shift mobility behaviour, as part of the Keli project (Kestävemmän liikkumisen kehittäminen hiilijalanjälkilaskurin avulla).
- › Project co-funded by the Ministry of the Environment. Results published in a [working paper](#).

Intern Embassy of Finland in Vienna

05/2021 - 08/2021

- › Monitored and reported on Austrian economic developments to inform Finnish Government policy decisions.
- › Attended and reported on meetings with UN organizations and local politicians.

// PROJECTS

Running Coach - Self-Hosted Training Planner

- › Building a self-hosted running coach that ingests Apple Health data into a canonical activity and vitals schema, maintains fitness and injury-load ledgers, and generates weekly plans from a deterministic guardrail engine.

Engineering: Docker, GitHub Actions, pytest, testthat
Web: Shiny, Quarto, PowerBI, REST APIs

// CERTIFICATIONS

- › **NVIDIA:** Fundamentals of Deep Learning
- › **NVIDIA:** Fundamentals of Accelerated Computing with CUDA Python

- › A Claude Agent SDK reviewer proposes plan changes that are validated against the guardrails before application. Python, FastAPI, SQLite, Docker; deployed on a home server behind Tailscale with pytest coverage across the engine.

cuplyr - GPU-Accelerated dplyr

- › Developing an R package that enables standard dplyr code to execute on GPU hardware through a RAPIDS cuDF backend.
- › Implements lazy evaluation with automatic query optimizations. Benchmarks show 40-77x speedups over dplyr on GPU-resident data; parity end-to-end including transfer.
- › [GitHub](#)

digitraffic - Fintraffic API Client

- › Building an R package for Finland's Digitraffic road traffic sensor time series: 450+ roadside sensors, per-vehicle records, real-time and historical.
- › Features tidyverse-native output, spatial filtering, built-in caching, and rate limiting. Tested with GitHub Actions; to be submitted to CRAN.
- › [GitHub](#)

bracketeer - Tournament Management Framework

- › Developed an R package for modeling and executing tournament competitions with support for multiple formats including round-robin, Swiss system, and elimination brackets.
- › Features a pipe-first API design for defining reusable tournament blueprints with automatic stage materialization and flexible result entry.
- › [GitHub](#)

Resepor - Recipe Assistant

- › [Python Shiny app](#) for interactive recipe creation on the Claude API via Chatlas.

// POSITIONS OF RESPONSIBILITY

Vote Counter - UNIDO

Authorized by the Western countries group to serve as vote counter in the Secretary-General Election of the United Nations Industrial Development Organization.

Board Member - Economics Students' Association

Served on the board contributing to strategic planning and student activities.